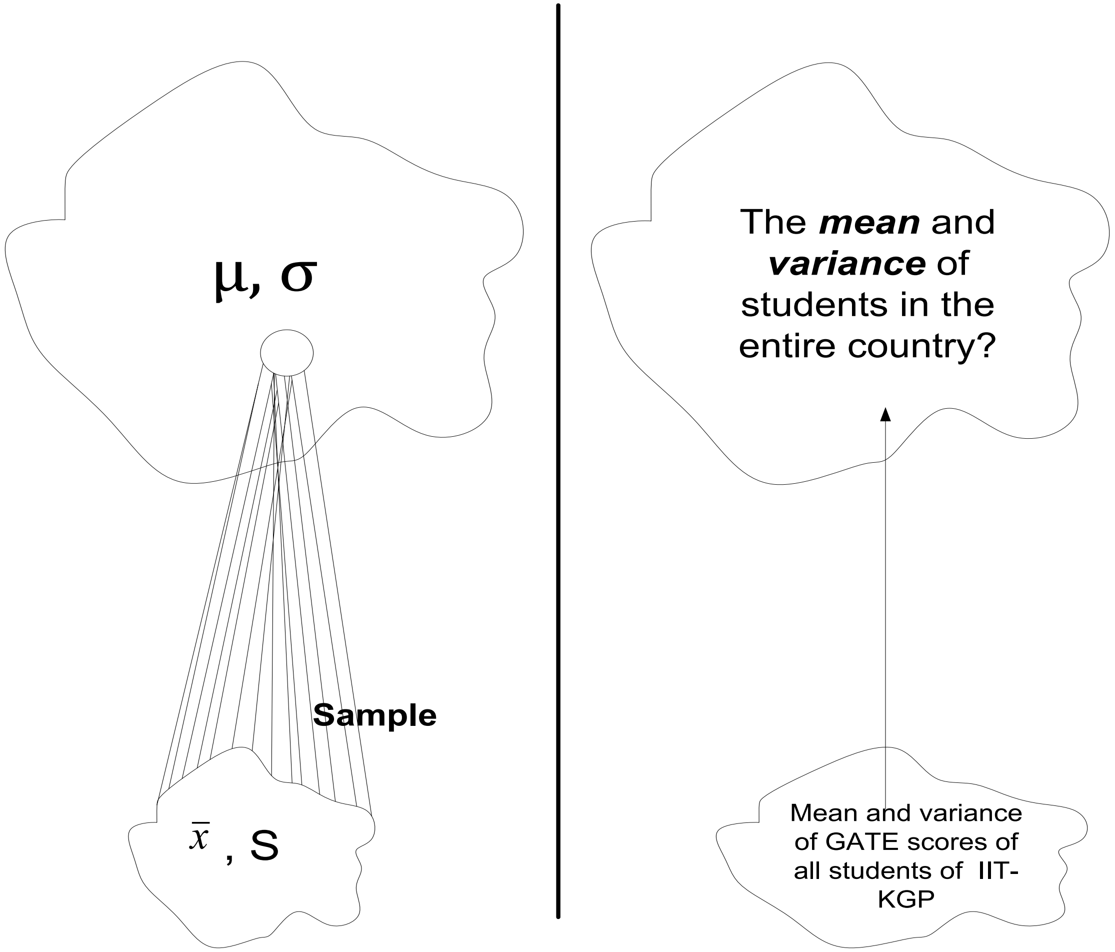
Unit 6

Testing of Hypothesis

The primary objective of statistical analysis is to use data from a sample to make inferences about the population from which the sample was drawn.



Hypothesis testing is a statistical method used to make decisions about population parameters based on sample data. It involves making an assumption (the hypothesis) about a population parameter, then using statistical tests to determine whether there is enough evidence to reject or fail to reject the assumption.

What is Hypothesis?

- "A hypothesis is an educated prediction that can be tested" (study.com)
- "A hypothesis is a proposed explanation for a phenomenon" (Wikipedia).
- "A hypothesis is used to define the relationship between two variables" (Oxford dictionary).

Statistical Hypothesis:

If the hypothesis is stated in terms of population parameters (such as mean and variance), the hypothesis is called a statistical hypothesis.

Example: **To determine whether the wages of men and women are equal.**

A product in the market is of standard quality.

Whether a particular medicine is effective to cure a disease.

Null and Alternate hypothesis

Null Hypothesis: It is a claim or statement about population parameter that is assumed to be true until it is declared to be false.

According to *Prof. R.A. Fisher,* null hypothesis is the hypothesis which is tested for the possible rejection under the assumption that it is true.

or

The statement that there is no effect or no difference (No Significant difference found), and it represents the default assumption. It is usually denoted by $\boldsymbol{H_0}$.

For example: if we want to test the null hypothesis that the population has specified mean μ_0 **(say), i.e.** $H_0: \mu = \mu_0$

Alternate hypothesis: Any hypothesis which is complimentary to null hypothesis is called an alternate hypothesis, usually denoted by H_1. The statement you want to test for. It represents an effect or difference (Significant difference found).

We have two kinds of alternative hypotheses:-

(a) One-sided alternative hypothesis

(b) Two-sided alternative hypothesis

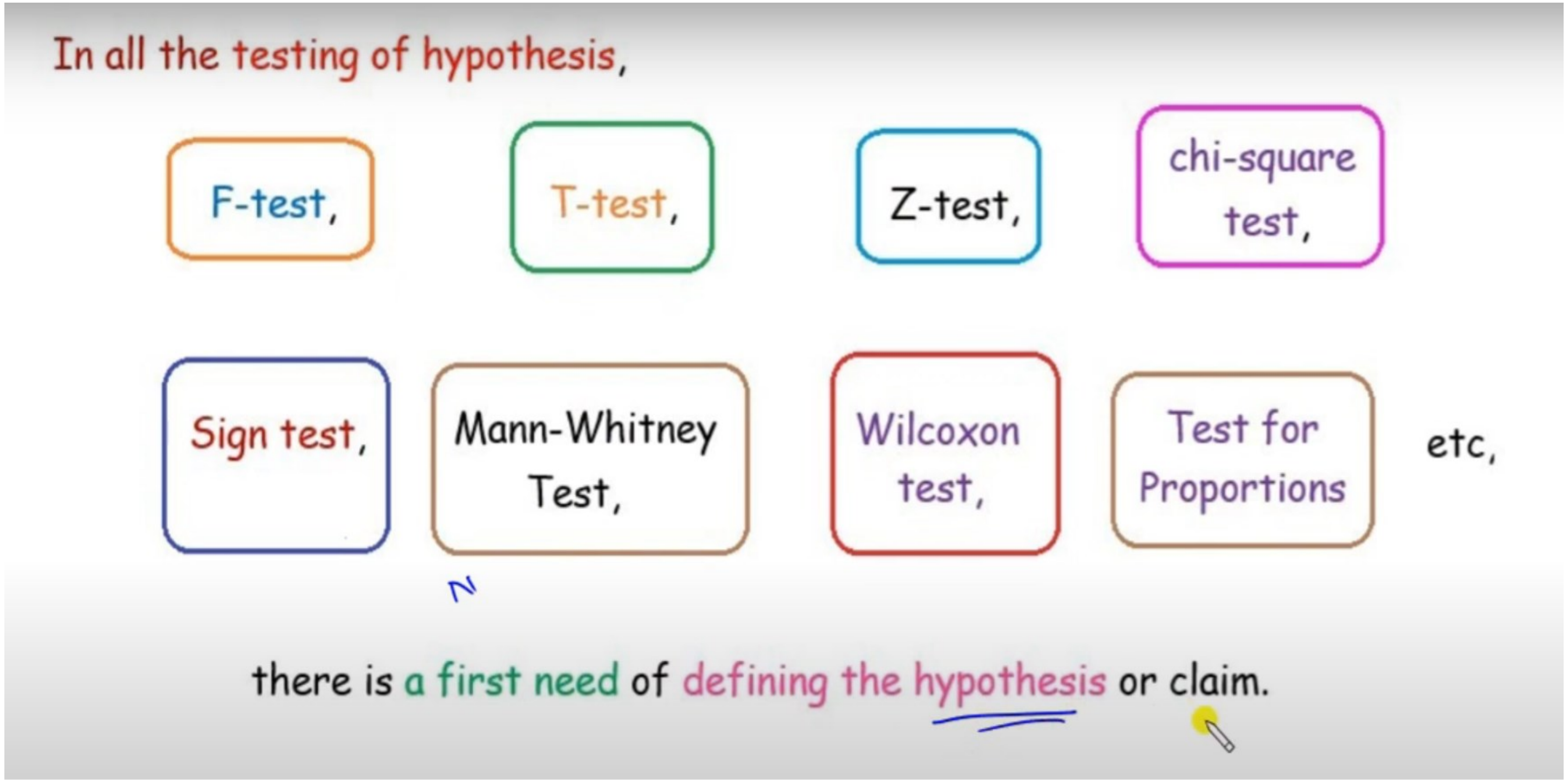
The test related to (a) is called a 'one-tailed' test and those related to (b) are called as 'two-tailed' tests.

Then the alternate hypothesis could be:

(i) $H_1: \mu \neq \mu_0$ *(two-tailed alternative)*

(ii) $H_1: \mu < \mu_0$ *(one tail left tail)*

(iii) $H_1: \mu > \mu_0$ *(one tail right tail)*



Possible outcomes?

Reject H0

Do not reject H0

When we have to reject or accept?

For rejection/acceptance of H0, we have to set the "Level of Significance"

Level of Significance: Probability that we make the error to reject the null hypothesis, when the null hypothesis is true.

Or

Level of risk in rejecting a correct Ho.

Or

The Level of Significance in hypothesis testing refers to the probability threshold set by the researcher to determine whether to reject the null hypothesis. The probability α that a random value of the statistic t belongs to the critical region is known as the level of significance. (level of significance is the size of type I error)

Common levels of significance include:

•**0.05** (5%): There's a 5% risk of concluding that the null hypothesis is false when it is actually true.
•**0.01** (1%): There's a 1% risk of making this error.
•**0.10** (10%): Used in some exploratory studies, with a higher tolerance for error.

Critical region and level of significance

A region corresponding to the statistic (t) in the sample space S which amounts to the rejection of H_0 is termed as critical region or region of rejection. If ω is the critical region and t is the value of the statistic based on the random sample of size n, then

$$P(t \in \omega | H_0) = \alpha, P(t \in \bar{\omega} | H_1) = \beta$$

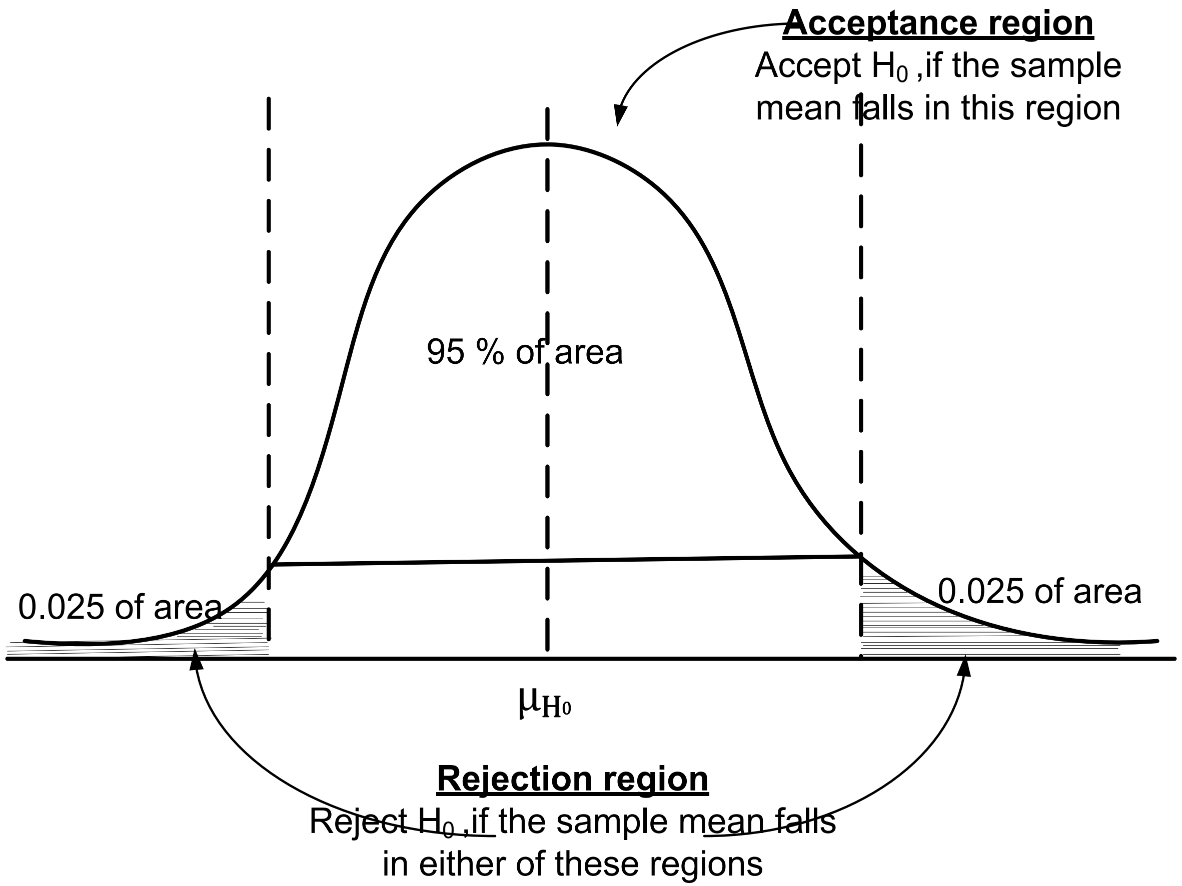
Where $\bar{\omega}$, the complementary set of ω, is called the acceptance region.

We have $\omega \cup \bar{\omega} = S$ and $\omega \cap \bar{\omega} = \phi$.

A critical value is any value that separates the critical region (where we reject the null hypothesis) from the values of the test statistic that do not lead to the rejection of the null hypothesis. The critical values depend on the nature of the null hypothesis, the sampling distribution that applies, and the significance level α.

Critical Region

The critical region (or rejection region) is the set of all values of the test statistic that cause us to reject the null hypothesis.



Acceptance and rejection regions in case of a two-tailed test with 5% significance level.

The significance level (denoted by α) is the probability that the test statistic will fall in the critical region when the null hypothesis is actually true. Common choices for α are 0.05, 0.01, and 0.10.

Critical values or significant values

- The value of the test statistic which separates the critical region and the acceptance region is called the critical value or significant value. It depends upon:

(i) The level of significance used and

(ii) The alternative hypothesis, whether it is two-tailed or one-tailed.

Critical Value α	Level of significance		
	1%	5%	10%
Two tailed test	$\lvert z_\alpha \rvert = 2.58$	$\lvert z_\alpha \rvert = 1.96$	$\lvert z_\alpha \rvert = 1.645$
Right tailed test	$z_\alpha = 2.33$	$z_\alpha = 1.645$	$z_\alpha = 1.28$
Left tailed test	$z_\alpha = -2.33$	$z_\alpha = -1.645$	$z_\alpha = -1.28$

Level of Confidence (or Confidence Level): It is typically expressed as a percentage and is complementary to the level of significance.

Mathematically, if the level of significance is **α**, then the confidence level is:

$$\text{Confidence Level} = (1-\alpha)\times 100\%$$

Common Confidence Levels:

- **90% Confidence Level** ($\alpha = 0.10$)
- **95% Confidence Level** ($\alpha = 0.05$)
- **99% Confidence Level** ($\alpha = 0.01$)

Interpretation:

A **95% confidence level** means that if we were to draw 100 different random samples and compute a confidence interval for each, we would expect 95 of those intervals to contain the true population parameter.

In hypothesis testing, the confidence level is related to how confident you are that the true parameter lies within a specific range (confidence interval) around the estimate.

Example:

- For a **95% confidence level**, we are 95% confident that the true population mean (or other parameter) falls within the calculated interval. This also means there is a 5% chance that the true parameter lies outside this range.

Which test do we have to apply? One-tailed and two-tailed test

Note: In any test, the critical region is represented by the portion of the area under the probability curve of the sampling distribution of the test statistic.

- **Two tailed test:** A test of the statistical hypothesis where the alternative hypothesis is two tailed such as $H_0: \mu = \mu_0$ against the alternative hypothesis $H_1: \mu \neq \mu_0$, is known as two tailed test and in such case the critical region is given by the portion of the area lying in both tails of the probability curve of the test statistic.
- **One tailed test:** A test of any statistical hypothesis where the alternative hypothesis is one tailed is called one tailed test.
- **For example :** Null Hypothesis: $H_0: \mu = \mu_0$ against the alternative hypothesis $H_1: \mu > \mu_0$(one tailed Right tailed)

or $H_1: \mu < \mu_0$ (one tailed left tailed)

In right tailed test, the critical region lies entirely in the right tail of the sampling distribution of $\bar{x}$, while for left tailed, the critical region is entirely in the left tail of the distribution.

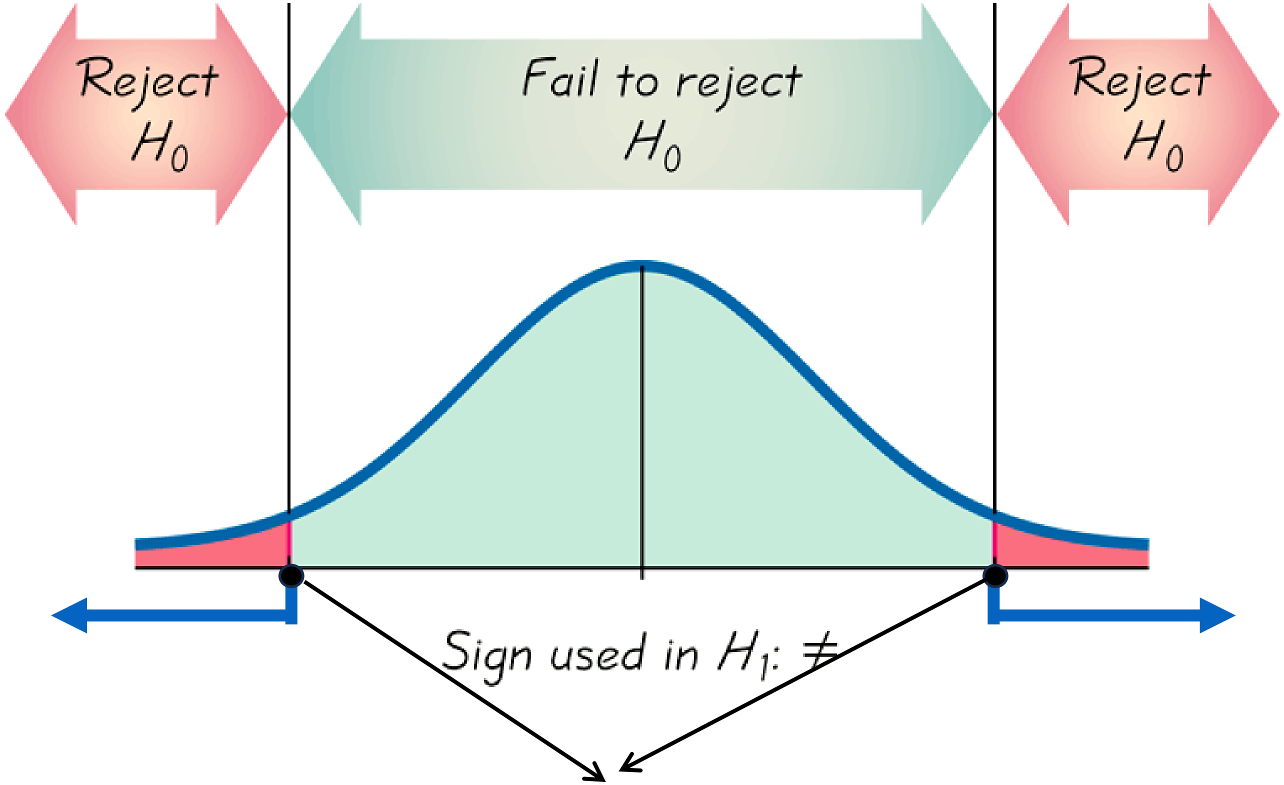
Two-tailed Test

H_0: $=$

H_1: $\neq$

Means less than or greater than

α **is divided equally between the two tails of the critical region**



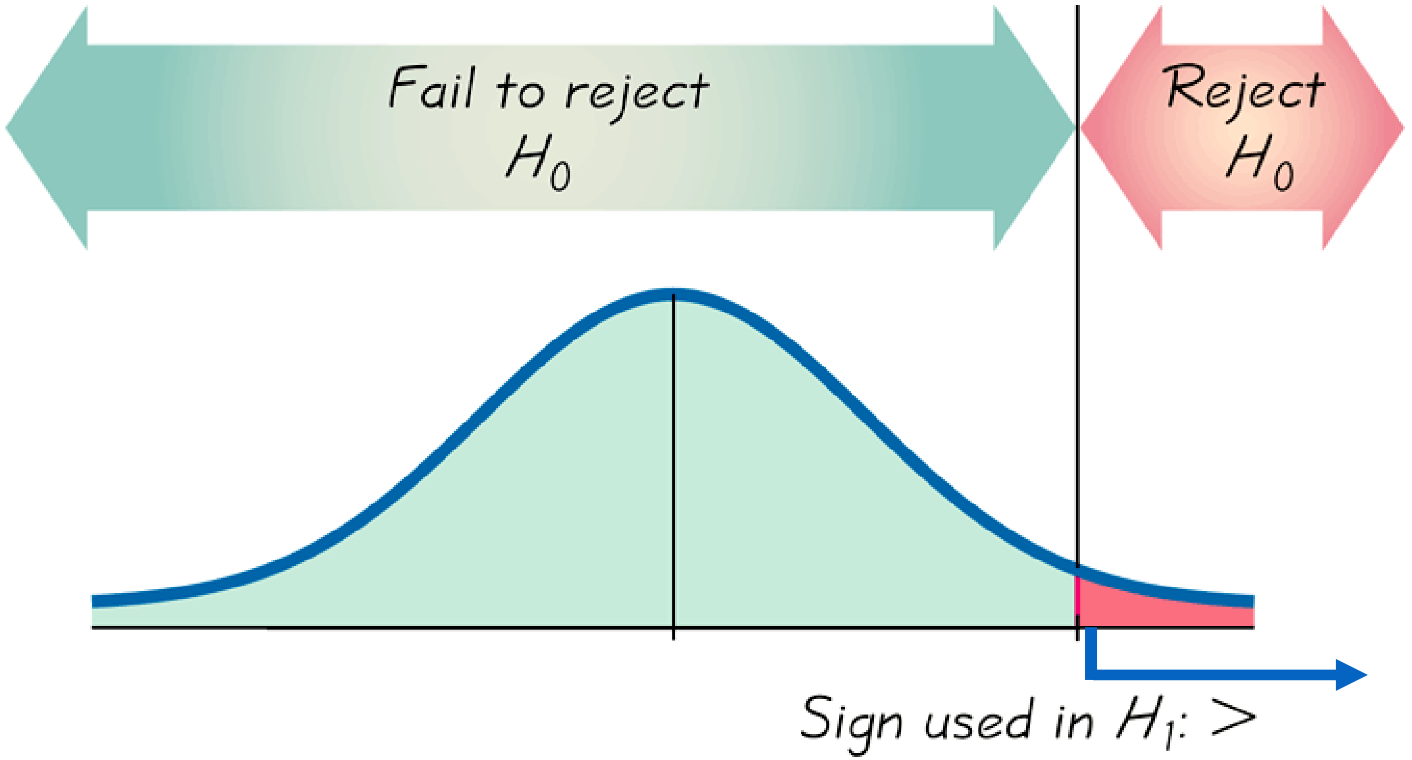
critical value

Right-tailed Test

H_0: =

H_1: >

Points Right

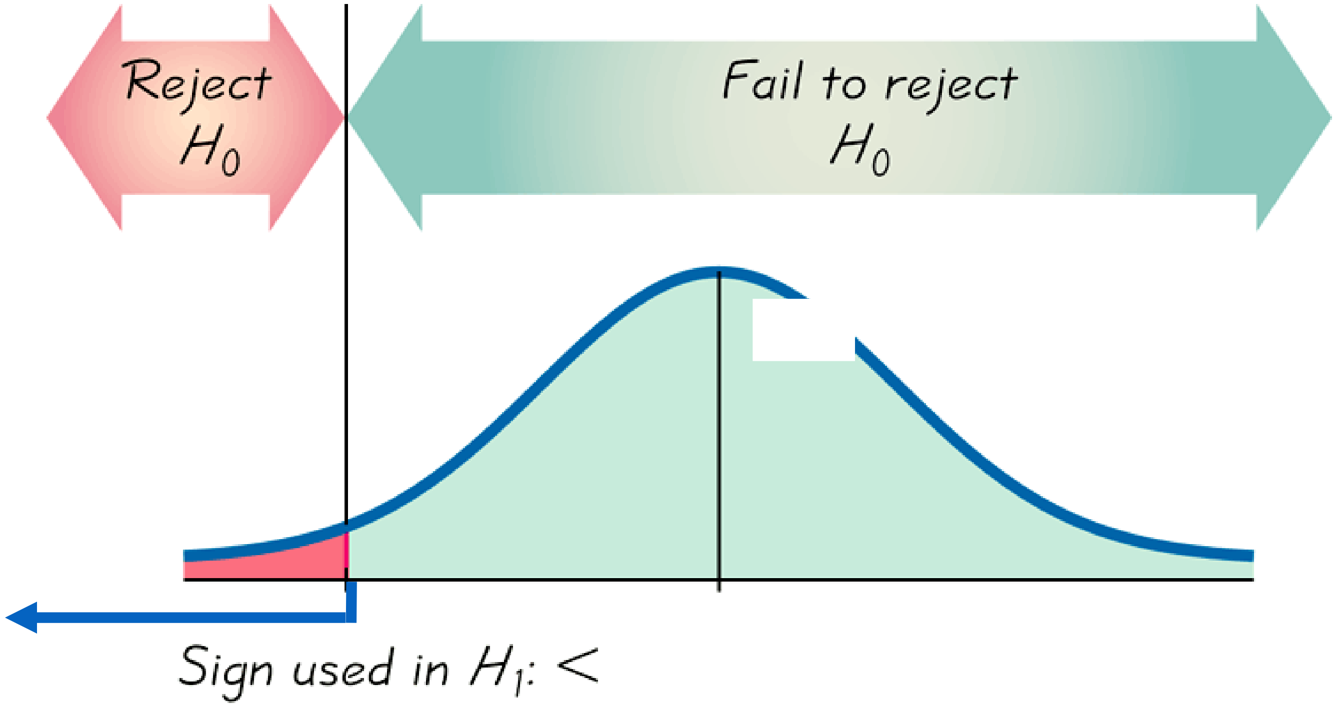


Left-tailed Test

H_0: =

H_1: <

Points Left



Decision:

If $|Z| < |Z_\alpha|$ ⇨ We accept H_0 (there is no significant difference between the sample statistics and population parameters)

Where $|Z|$ is calculated value and $|z_\alpha|$ is tabular value of Z at α level of significance

If $|Z| > |Z_\alpha|$ ⇨ We reject H_0 (there is a significant difference between the sample statistics and population parameters)

Type 1 and Type 2 Error

When we are testing Null hypothesis (H_0) against alternative hypothesis (H1) there are 4 possibilities.

H_0 accepted when H0 is true [correct decision]
H_0 is rejected when H0 is true [Type 1 error]
H_0 accepted when H0 is false [Type 2 error]
H_0 is rejected when H0 is false [correct decision]

Decision	Reality	
	H0 is true	H0 is false
Reject H0	Type I error	Correct decision
Fail to reject H0	Correct decision	Type II error

Type 1 error	Type 2 error
$\alpha = P(\text{Type 1 error})$	$\beta = P(\text{Type 2 error})$
$\alpha = P(\text{Reject } H_0 / H_0 \text{ is true})$	$\beta = P(Accept\ H_0 / H_1\ is\ true)$

Then α and β are called the sizes of type I and type II error.
The power of a test is the probability of rejecting H_0 given that a specific alternative is true.
The power of a test can be computed as $\mathbf{1 - \beta}$.

Basic definitions and terminology

- **Standard error:** The standard deviation of the sampling distribution of statistic is called standard error.

- **Utility of standard error:** If the discrepancy between the observed and the expected value of a statistic is **greater than z_α times the standard error**, the null hypothesis is rejected at α level of significance.

Procedure for testing of Hypothesis

Z-test for single mean

Apply Z-test when sample size $n \geq 30$.

Apply t-test when sample size $n < 30$.

Process :

1) Define Null and Alternative hypothesis
2) Test Statistics $z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$ [σ is known]

   $z = \frac{\bar{x}-\mu}{s/\sqrt{n}} \sim N(0,1)$ [σ is unknown]

3) Conclusion

If calculated $|z| > z_\alpha$,Null hypothesis is rejected

If calculated $|z| < z_\alpha$,Null hypothesis is accepted

Problems

Q1 : A sample of 900 members has a mean 3.4. cms. And standard deviation s.d. 2.61 cms. Is the sample from a large population of mean 3.25 cms. and s.d. 2.61.cms. If the population is normal and its mean is unknown, find the 95% and 98% fiducial limits of true mean.

Practice Questions

Q1 The mean muscular endurance score of a random sample of 60 subjects was found to be 145 with s.d. of 40. Construct a 95% confidence interval for the true mean. Assume the sample size to be large enough for normal approximation. What size of sample is required to estimate the mean within 5 of the true mean with 95% confidence.

Q2: An insurance agent has claimed that the average age of policy holders who insure through him is less than the average for all agents, which is 30.5 years. A random sample of 100 policy holders who had insured through him gave the following age distribution. Use that data to test his claim at 5% level of significance.

Age last birthday	**No. of persons**			
16-20	12			
21-25	22			
26-30	20			
31-35	30			
36-40	16			

Test of significance for difference of means

12·14. Test of Significance for Difference of Means. Let $\bar{x}_1$ be the mean of a random sample of size n_1 from a population with mean μ_1 and variance σ_1^2 and let $\bar{x}_2$ be the mean of an independent random sample of size n_2 from another population with mean μ_2 and variance σ_2^2. Then, since sample sizes are large,

$$\bar{x}_1 \sim N(\mu_1, \sigma_1^2/n_1) \text{ and } \bar{x}_2 \sim N(\mu_2, \sigma_2^2/n_2)$$

Also $\bar{x}_1 - \bar{x}_2$, being the difference of two independent normal variates is also a normal variate. The Z ($S.N.V.$) corresponding to $\bar{x}_1 - \bar{x}_2$ is given by

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - E(\bar{x}_1 - \bar{x}_2)}{S.E.(\bar{x}_1 - \bar{x}_2)} \sim N(0, 1)$$

Under the null hypothesis $H_0 : \mu_1 = \mu_2$, *i.e.*, there is no significant difference between the sample means, we get

$$E(\bar{x}_1 - \bar{x}_2) = E(\bar{x}_1) - E(\bar{x}_2) = \mu_1 - \mu_2 = 0;$$

$$V(\bar{x}_1 - \bar{x}_2) = V(\bar{x}_1) + V(\bar{x}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2},$$

the covariance term vanishes, since the sample means $\bar{x}_1$ and $\bar{x}_2$ are independent.

Thus under $H_0 : \mu_1 = \mu_2$, the test statistic becomes (for large samples),

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{(\sigma_1^2/n_1) + (\sigma_2^2/n_2)}} \sim N(0, 1) \qquad \ldots(12{\cdot}11)$$

When Variance of population is known

Remarks 1. If $\sigma_1^2 = \sigma_2^2 = \sigma^2$, *i.e.*, if the samples have been drawn from the populations with common S.D. σ, then under $H_0 : \mu_1 = \mu_2$,

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{(1/n_1 + 1/n_2)}} \sim N(0, 1) \qquad \ldots[12\cdot11(a)]$$

When Variance of population is not known

2. If in (12·11a), σ is not known, then its estimate based on the sample variances is used. If the sample sizes are not sufficiently large, then an unbiased estimate of σ^2 is given by

$$\hat{\sigma}^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 + n_2 - 2)},$$

since

$$E(\hat{\sigma}^2) = \frac{1}{n_1 + n_2 - 2}\left[(n_1 - 1)\,E(S_1^2) + (n_2 - 1)\,E(S_2^2)\right]$$

$$= \frac{1}{n_1 + n_2 - 2}\left[(n_1 - 1)\,\sigma^2 + (n_2 - 1)\,\sigma^2\right] = \sigma^2$$

But since sample sizes are large, $S_1^2 \simeq s_1^2$, $S_2^2 \simeq s_2^2$, $n_1 - 1 \simeq n_1$, $n_2 - 1 \simeq n_2$. Therefore in practice, for large samples, the following estimate of σ^2 without any serious error is used :

$$\hat{\sigma}^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2} \qquad \text{...[12·11}(b)\text{]}$$

However, if sample sizes are small, then a small sample test, t-test for difference of means (*c.f.* Chapter 14) is to be used.

3. If $\sigma_1^2 \neq \sigma_2^2$ and σ_1 and σ_2 are not known, then they are estimated from sample values. This results in some error, which is practically immaterial, if samples are large. These estimates for large samples are given by

$$\left.\begin{aligned} \hat{\sigma}_1^2 &= S_1^2 \simeq s_1^2 \\ \hat{\sigma}_2^2 &= S_2^2 \simeq s_2^2 \end{aligned}\right\} \qquad \text{(since samples are large).}$$

In this case, (12·11) gives

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{(s_1^2/n_1) + (s_2^2/n_2)}} \sim N(0,\, 1) \qquad \text{...[12·11}(c)\text{]}$$

1. The means of two single large samples of 1000 and 2000 members are 67·5 inches and 68·0 inches respectively. Can the samples be regarded as drawn from the same population of standard deviation 2·5 inches? (Test at 5% level of significance).

2. In a survey of buying habits, 400 women shoppers are chosen at random in super market 'A' located in a certain section of the city. Their average weekly food expenditure is Rs. 250 with a standard deviation of Rs. 40. For 400 women shoppers chosen at random in super market 'B' in other section of the city, the average weekly food expenditure is Rs. 220 with a standard deviation of Rs. 55. Test at 1% level of significance whether the average weekly food expenditure of the two populations of shoppers are equal.

3. The average hourly wage of a sample of 150 workers in a plant 'A' was Rs. 2·56 with a standard deviation of Rs. 1·08. The average wage of a sample of 200 workers in plant 'B' was Rs. 2·87 with a standard deviation of Rs. 1·28. Can an .applicant safely assume that the hourly wages paid by plant 'B' are higher than those paid by plant ‘A' ?

Practice Questions

Q1: The means of two single large samples of 1000 and 2000 members are 67.5 inches and 68 inches respectively. Can thee sample be regarded as drawn from the same population of standard deviation 2.5 inches?

Example 12·26. *In a certain factory there are two independent processes manufacturing the same item. The average weight in a sample of 250 items produced from one process is found to be 120 ozs. with a standard deviation of 12 ozs. while the corresponding figures in a sample of 400 items from the other process are 124 and 14. Obtain the standard error of difference between the two sample means. Is this difference significant ? Also find the 99% confidence limits for the difference in the average weights of items produced by the two processes respectively.*

Q3 : The average hourly wage of a sample of 150 workers in a plant *A* was Rs. 2.56 with a standard deviation of Rs. 1.08. The average hourly wage of a sample of 200 workers in plant *B* was Rs. 2.87 with a standard deviation of Rs. 1.28. Can an applicant safely assume that th hourly wages paid by plant *B* are higher than those paid by plant *A*?

Practice Questions

Q4: In a certain factory there are two independent processes manufacturing the same item. The average weight in a sample of 250 items produced from one process is found to be 120 ozs. With a standard deviation of 12 ozs. While the corresponding figures in a sample of 400 items from the other process are 124 and 14. Obtain the standard error of difference between the sample means. Is this difference significant? Also find the 99% confidence limits for the difference in the average weights of items produced by the two processes respectively.

Example 12·24. *In a survey of buying habits, 400 women shoppers are chosen at random in super market 'A' located in a certain section of the city. Their average weekly food expenditure is Rs. 250 with a standard deviation of Rs. 40. For 400 women shoppers chosen at random in super market 'B' in another section of the city, the average weekly food expenditure is Rs. 220 with a standard deviation of Rs. 55. Test at 1% level of significance whether the average weekly food expenditure of the two populations of shoppers are equal.*

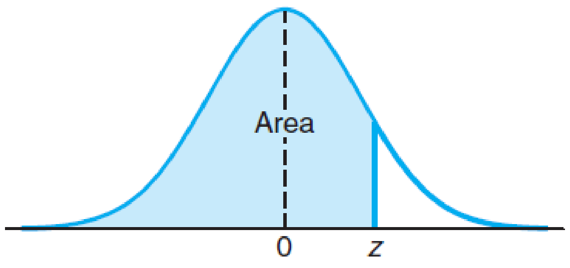


Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143

−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Table A.3 (continued) Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998